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Smart AI Baby Body Scanner (SABBS): A Mobile Application for Infant Height Measurement

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Abstract. Infant growth is an important indicator of their health and development, making accurate and regular measurements essential. However, traditional measurement methods are often inaccurate and require frequent visits to healthcare centers. In this report, we developed a infant height measurement application using smartphone camera technology and image processing algorithms to provide a practical and accurate solution. The application not only simplifies height measurement but also facilitates the recording and analysis of infant growth data. Parents can store measurement history digitally and share this data with doctors or healthcare professionals for a more comprehensive assessment. This feature is especially useful for the early detection of potential growth issues, such as stunting, allowing for timely medical intervention. Testing shows that the application has varying degrees of accuracy, with some subjects demonstrating high accuracy while others require further adjustments for consistent results.

Introduction

Infant growth is an important indicator in assessing their health and development, so precise and regular measurements are needed [1]. The infant's height is usually measured using a height ruler, with two medical staff members working together to hold the infant's head and feet steady during the measurement [2]. However, parents often face obstacles in carrying out these measurements at home with traditional methods that are less accurate or require regular visits to the health center which are time consuming. With advances in technology, infant height measurement applications offer a practical solution by using smartphones using cell phone camera technology and image processing algorithms to measure infant height. In addition to ease and accuracy of measurement, the application also simplifies the recording and analysis of infant growth data. With the app, parents can digitally store the infant's height measurement history, making it easier to track the child's development over time. This data can also be easily shared with doctors or other healthcare professionals when needed, aiding in providing a more comprehensive health assessment [3]. This feature is especially useful for early detection of potential growth problems, such as stunting, and allows for timely medical intervention.

Related Works

Infant height estimation has transitioned from traditional contact-based methods to advanced computer vision and machine learning techniques, offering more efficient, accurate, and contactless solutions. Early approaches, such as those employing ultrasonic ranging [4], measured height by calculating the distance between two points using ultrasonic waves. While this method automates the process, it is limited by mechanical components and sensor dependency. Similarly, Kinect-based methods [5] used image processing to estimate the height of infants by extracting key points along the body centerline. Though effective, this method faced challenges with camera motion and image

instability, which required filtering techniques to improve accuracy. Recent advancements have introduced deep learning models, significantly improving the robustness and precision of height estimation. Systems like the one proposed by [6] integrated OpenPose for body key-point detection with DeepLabV3+ for image segmentation, providing a fully noncontact approach that optimizes feature extraction for more accurate height measurements. Similarly, [2] applied lightweight convolutional networks like MobileNet to estimate infant height and weight from images, achieving a mean absolute error of less than 1.1 cm. In another approach, [7] utilized depth images collected via smartphones to estimate the height of standing children, further demonstrating the effectiveness of CNNs in measuring height accurately even in resource-constrained environments.

System Design and Implementation

Application Feature Design. The application that was built has the main feature, namely being able to calculate the infant's height based on photos and being able to save the measurement results on the device and displaying a graph of the history of measurements that have been taken. To implement this feature, the application must meet the following requirements.

a) Functional requirements, which define core capabilities of the application as follows.
   - The application can take pictures and send them to the server.
   - The application can display the results of calculating the infant's height based on the image taken.
   - The server can receive images from the application.
   - The server can process the image and produce an estimate of the infant's height.
   - The server can send calculation results to the application.
   - The application can save calculation results and their time to a local database.
   - The application can display calculation history from a local database in graphical form.

b) System requirements, which ensure that the application has necessary system specifications as follows.
   - The application requires phone camera access permission to take infant pictures.
   - The application must be compatible with commonly used mobile operating systems such as Android or iOS.
   - The application requires an internet connection to be able to upload images to the server.

In accordance with these requirements, the application performance flow design is presented using the activity diagram shown in Fig. 1.

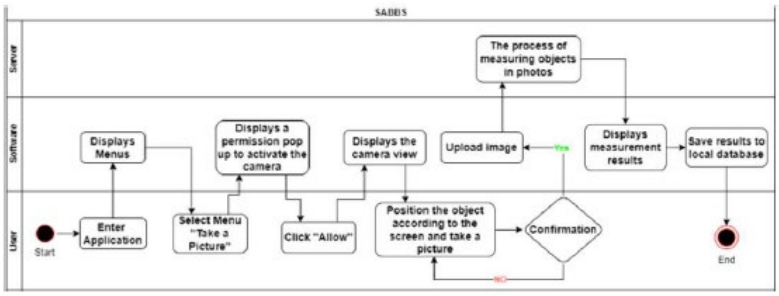

Fig. 1. Activity diagram.

Height Measurement Methods. Height measurement based on image processing is implemented using the Python language by utilizing the Media Pipe [8, 9] and OpenCV modules [10]. The module provides libraries and tools to quickly apply AI and Machine Learning techniques in applications. Media Pipe provides the Media Pipe Pose Land marker submodule which makes it possible to detect 33 human body landmarks in images or videos [9]. Illustrations of landmarks and their descriptions are presented in Fig. 2 and Fig. 3, respectively.

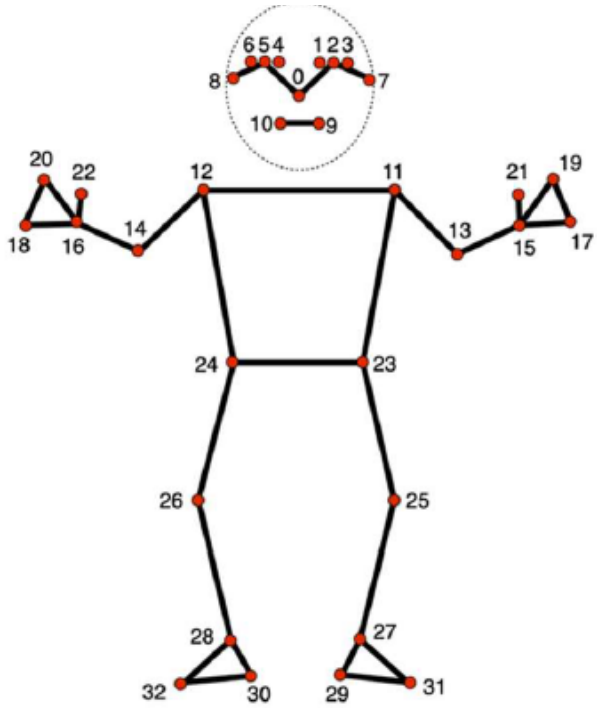

Fig. 2. Illustration and position of landmarks.

0 - nose	17 - left pinky
1 - left eye (inner)	18 - right pinky
2 - left eye	19 - left index
3 - left eye (outer)	20 - right index
4 - right eye (inner)	21 - left thumb
5 - right eye	22 - right thumb
6 - right eye (outer)	23 - left hip
7 - left ear	24 - right hip
8 - right ear	25 - left knee
9 - mouth (left)	26 - right knee
10 - mouth (right)	27 - left ankle
11 - left shoulder	28 - right ankle
12 - right shoulder	29 - left heel
13 - left elbow	30 - right heel
14 - right elbow	31 - left foot index
15 - left wrist	32 - right foot index
16 - right wrist	

Fig. 3. Description of each landmark.

Not all body landmark points are connected and used in the process, necessitating calibration for accurate infant height measurement. By adding a new point (33) to identify the location of the top of the head, the points utilized for measuring infant height are 30 - 28 - 26 - 24 - 12 - 5 - 33. The addition of this new point follows the process outlined below.

1. The landmarks on the shoulders, specifically points 11 and 12, are brought closer by applying dilation to the vector connecting the two points, with the center of dilation being the midpoint between them, and the dilation factor set at 0.6.
2. A new point (33) is generated, such that the vector from point 0 to 33 has the same direction as the vector from point 24 to 12, but its length is 0.925 times that of the latter.

Consequently, the infant's height, expressed in pixels, is determined by summing the lengths of the vectors connecting the selected points. To obtain the height in real-world units, a reference object is required. Open CV provides a function to segmentate based on HSV color ranges and detect contours. A contour can be explained simply as a line connecting all continuous points (along a boundary), which have the same color or intensity. To detect reference objects, the color value limits used are in the range of HSV (30, 40, 40) and HSV (90, 255, 255) values, which represent green color. After performing HSV-based color segmentation, a binary image containing the reference object is obtained, allowing contour detection to be performed on the object. The detection result is a bounding box that forms a reference object, so that the length of the bounding box can be used for measurement. Finally, if the length of the infant's vector in pixels is denoted as x, the length of the reference object's vector in the bounding box is y, and the actual length of the reference object is l cm, then the infant's height is calculated as x/y cm.

Server API Design. The height measurement module is integrated into the backend system for image processing, where the backend is implemented using Python, leveraging the Flask framework [11]. The API is used to handle height measurement requests from the application to the server and returns image processing results in the form of annotated images and height values from the server to the application. The design of the developed API is presented in Table I.

Table I. Api Design.

Endpoint	/detect
Method	POST
Request Header	• Accept : multipart/form-data
Format Formdata	• photo (file) : infant image file (.png, .jpg, .jpeg) • refLength (float) : reference object length in cm
Response Header	• Accept : application/json
Format Response	• annotation (string) : base 64 encoded image with annotation (landmark line) • height (float) : infant height measurement result in cm

Application System Design. The application system design includes main components such as user interface as well as database design.

a) User interface (UI/UX). Wireframes and mockups are created to ensure the user interface is intuitive and easy to use. User Interface (UI/UX) design was carried out using Figma. Figma was chosen as the primary tool because of its ability to support real-time collaboration, interactive prototyping, and responsive design. The following is the page design that will be implemented into the application.
   - Splashscreen page: initial page when the application is first run. On this page there is a process in the background to prepare the system for the application.
   - Landing page: play an important role in providing a good first impression and setting user expectations for the application, ensuring users feel interested and ready to explore the features on offer.
   - Home page: displays a list of infant profiles and can add new infant profiles.

- Dashboard page: displays the infant's profile, a button to calculate the infant's height and a button to display the infant's measurement history.
- Infant profile page: displays the infant's profile in detail and can be edited.
- Calculate height page: displays the steps that need to be taken to measure the infant's height with visual instructions.
- Measurement history page: displays graphs and lists showing the history of the infant's height from time to time.

b) Database design. Database is utilized to store profile information for infants undergoing measurement, including their name, date of birth, and gender, as well as measurement history, which records past height measurements along with their respective timestamps. Each infant profile entity can be associated with multiple measurement history entities. The class diagram and data flow diagram are presented in Fig. 4 and 5, respectively, while the data dictionary for each entity is shown in Fig. 6 and 7.

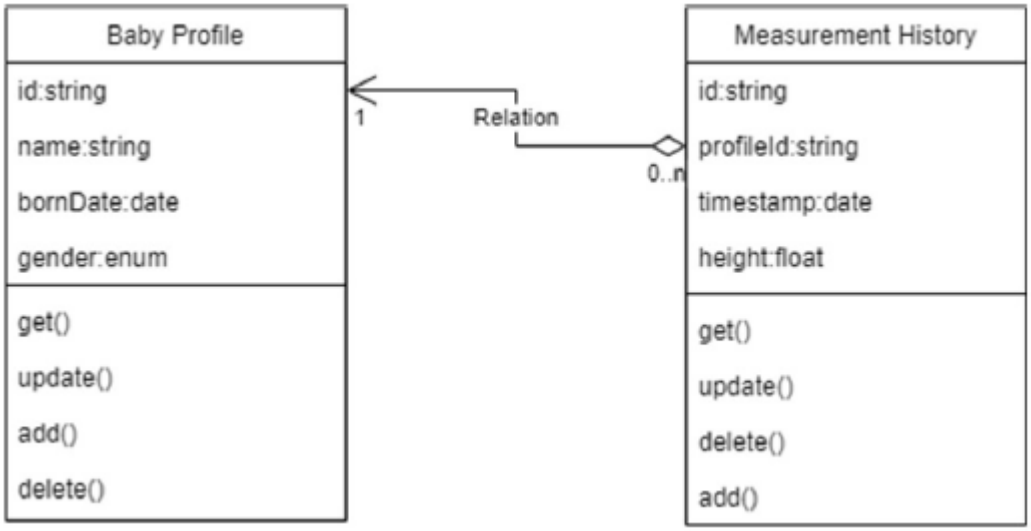

Fig. 4. Class diagram.

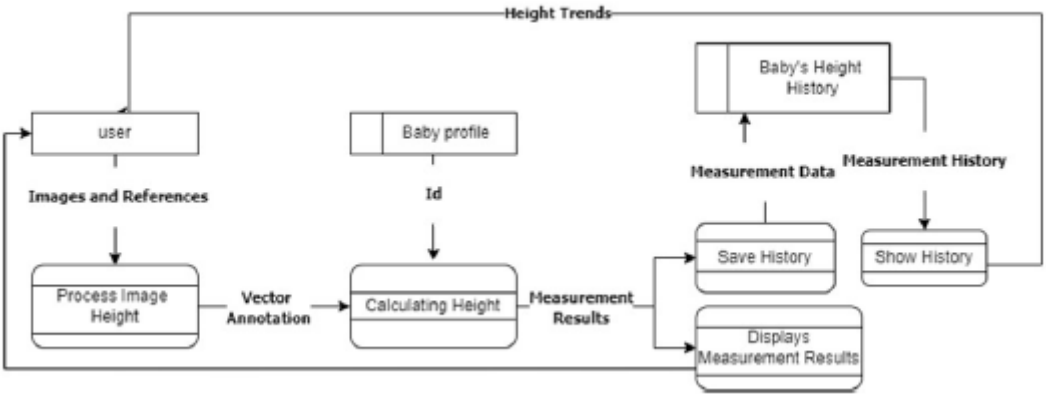

Fig. 5. Data flow diagram.

Column	Description	Type	Length	Primary Key	Nullable	Unique
id	Unique identification	String	255	True	False	True
name	Baby names	String	255	False	False	False
bornDate	Baby's date of birth in the format dd-mm-yyyy	String	10	False	False	False
gender	Baby's gender in the format male (M) and female (F)	String	1	False	False	False

Fig. 6. Data dictionary for infant profile.

Column	Description	Type	Length	Primary Key	Nullable	Unique
id	Unique identifier	String	255	True	False	True
profileId	ID of the baby profile being measured	String	255	False	False	False
timestamp	Date and time of measurement in the format (YYYY-MM-DD hh:mm:ss)	String	20	False	False	False
height	Measurement result in cm	Float	5	False	False	False

Fig. 7. Data dictionary for measurement history.

Application Architecture Design. The application architecture consists of three layers: the Data layer (comprising components such as models, data sources, and repositories), the Domain layer (containing entities and use cases), and the Presentation layer (which includes the logic holder and widgets) [12]. The Presentation layer follows the Model-View-Controller (MVC) design pattern [13]. The architecture flow is illustrated in Fig. 8.

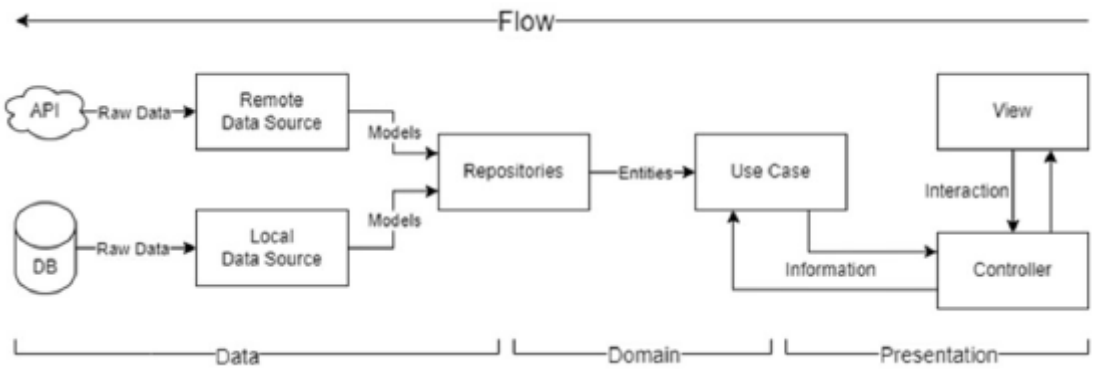

Fig. 8. Architecture flow.

The application is implemented using the Flutter framework [14], which utilizes the Dart programming language. This framework enables the development of mobile, web, and desktop applications from a single codebase, meaning the code only needs to be written once for deployment across multiple platforms. The target platform for this system's development is mobile (Android and iOS), with initial testing specifically conducted on Android devices.

The presentation layer employs the GetX state management for implementing the design pattern. Dependency injection, using the singleton pattern, is applied to manage dependencies across objects in the various layers [15]. The local database used is Hive, a NoSQL database [16].

Development Methodology. The development of the infant height measurement application follows the Agile methodology [17], which enables iterative and continuous improvement through incremental development and constant feedback. The first stage involves requirements analysis and system design, as described in the previous sections. The next phase is testing, conducted to ensure the application functions according to the specified requirements. Unit testing is performed to evaluate each component of the application individually, while integration testing ensures seamless workflow between the front-end, back-end, and machine learning modules. System testing is conducted end-to-end, and user testing is performed to gather feedback for further improvement. After successful testing, the application is launched in a beta phase to a limited group of users to gather initial feedback. The infant height measurement application is expected to be developed effectively, meet user needs, and remain relevant through continuous updates and improvements.

Result And Analysis

Application System Implementation. The interface design was created using Figma which can be viewed at the link https://bit.ly/sabbs-ui-design and application source code repository can be accessed with the link https://github.com/Project-PPL-MKOM/ppl-fe, while the developed application can be accessed and downloaded from the link https://bit.ly/sabbs-apk. The application was tested using both several real devices and an android simulator to check its interface and inspect whether the system works properly or not. An example of one of the application's pages is shown in Fig. 9.

Fig. 9. Example of a height calculation page display.

In the implementation phase of the height measurement system, the landmark detection results are shown in the image with indicators in the form of circles and connecting lines representing vector

components for calculation. Meanwhile, the detection results on the reference object are shown in the image outlined by a rectangular frame. An example of landmark detection results and reference objects is shown in Fig. 10.

Fig. 10. Example of landmark and reference object detection results.

API Unit Testing. In API testing, a series of unit tests were conducted to ensure that the API provides appropriate responses and status codes for each request scenario. The results of the testing conducted using Postman are shown in Fig. 11. Based on this figure, it can be seen that the API implementation successfully passed all test cases. Backend and API source code repository can be accessed with the link https://github.com/Project-PPLMKOM/ppl-be-api.

Fig. 11. Backend testing results.

Height Measurement Evaluation. For this application testing, 3 subjects were used: a 4-year-old child (Subject 1) with an actual height of 101 cm, a 10-month-old infant (Subject 2) with an actual height of 70.3 cm, and a baby mannequin (Subject 3) with an actual height of 37 cm. To conduct the test, 30 images of each subject were taken in a various angle and pose, with a 30.5 cm ruler as reference object included in each shot for scale. To ensure a green reference object, the ruler will be fully covered with green paper. The test results are shown in Table II.

Table II. Test Results On Each Subjects.

Subject	MAE (cm)	MAPE (%)
1	1.489	1.454
2	8.539	12.147
3	2.696	7.287

Table II presents the mean absolute error (MAE) and mean absolute percentage error (MAPE) for each subject. The errors were calculated by comparing each test result to the average percentage for each subject. The results indicate significant variations in measurement accuracy across subjects. Subject 1 demonstrated high accuracy with MAE 1.489 and MAPE 1.454% which is very low. However, Subject 2 exhibited lowest accuracy with MAE 8.539 and MAPE 12.147% due to imperfect reference object detection in some samples. Subject 3 with higher MAE and MAPE than Subject 1 also encountered challenges, with non-ideal body poses occasionally leading to overestimated vector lengths. These findings highlight the need for further evaluation and adjustments to ensure consistent and accurate measurement results for all subjects.

The overall MAE and MAPE is 4.241 and 6.963%, suggesting that the variability is significant for two subjects, which could point to inconsistencies or outliers in the data for those individuals. The next test was conducted using 30 images only subject 3, but with variations in the length of the reference object provided. The evaluation MAE and MAPE from scenario are shown in Fig. 12 and Fig. 13, respectively

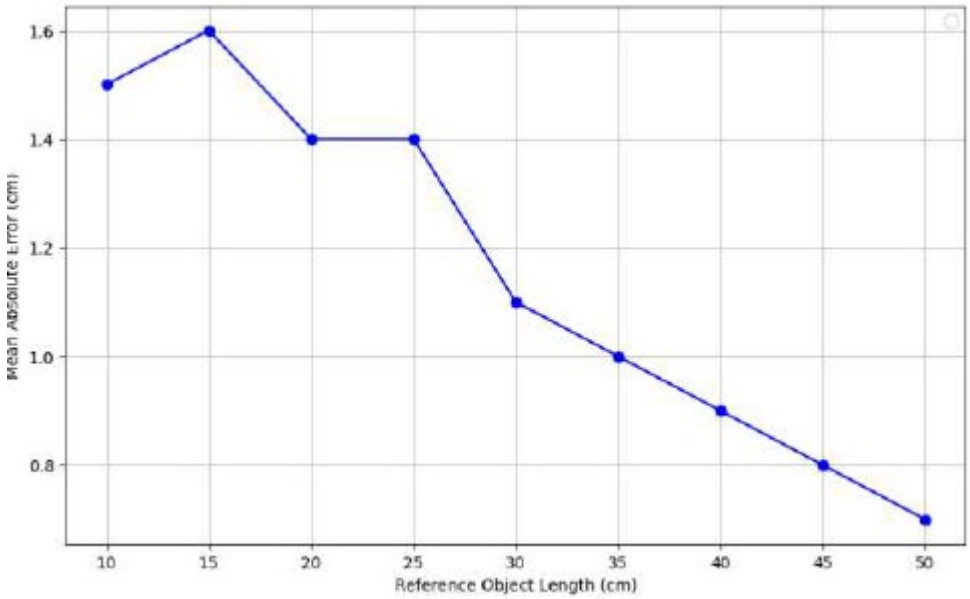

Fig. 12. Evaluation MAE with varying reference object length.

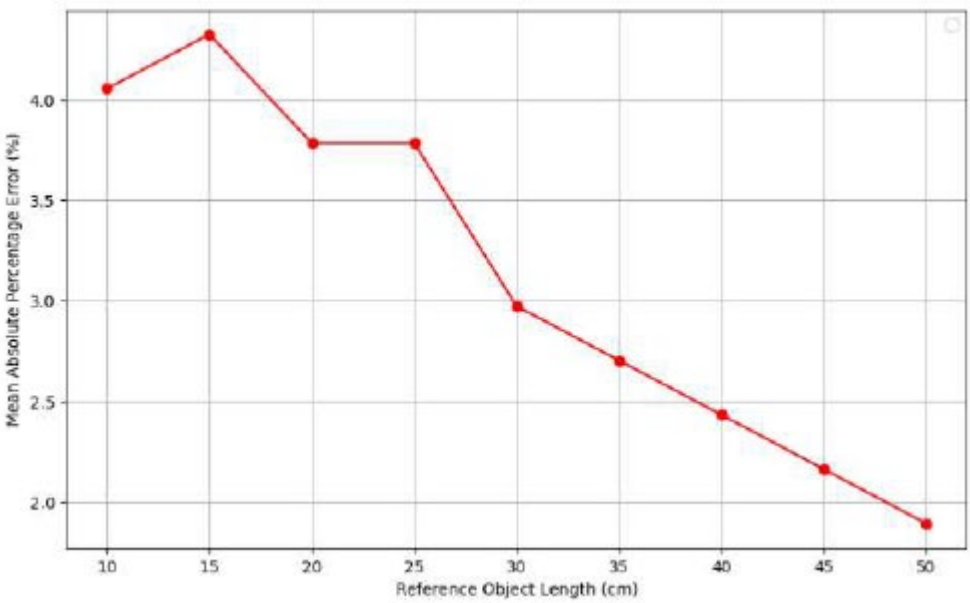

Fig. 13. Evaluation MAE with varying reference object length.

The graph above shows that longer reference objects generally lead to more accurate height measurements. However, the improvement in accuracy is not always significant, especially for reference objects shorter than 15 cm. Factors like image quality and object complexity can also influence measurement accuracy. Therefore, optimizing these factors in addition to using appropriate reference object lengths is crucial for achieving highly accurate height measurements.

Conclusion

The height measurement application was developed and tested successfully using a combination of real devices and an Android simulator to verify its functionality and user interface. The application interface, created in Figma, as well as the source code repositories for both frontend and backend, were made accessible for further evaluation. During testing, the system's ability to detect landmarks and reference objects was visually confirmed, with the results indicating the accuracy of body height estimation. The proposed machine learning model demonstrates the ability to measure body height with relatively high accuracy, particularly under ideal conditions where the subject maintains an upright posture and a clear, uniformly green reference object is used. Subject 1, for example, achieved a low Mean Absolute Error (MAE) of 1.489 cm and a Mean Absolute Percentage Error (MAPE) of 1.454%, indicating strong performance in favorable scenarios. However, the model exhibited reduced accuracy in less optimal conditions, such as with Subject 2 (MAE 8.539 cm, MAPE 12.147%), where reference object detection challenges led to higher errors. Similarly, non-ideal body poses and variations in the reference object length affected the precision of Subject 3's measurements. Overall, the system achieved an MAE of 4.241 cm and a MAPE of 6.963%, but further improvements are necessary to ensure consistent accuracy across different subjects and conditions.

Suggestions For Future Work

Based on the findings, the following recommendations are made to enhance the accuracy and robustness of the height measurement system.

- To improve measurement consistency, using a longer reference object (e.g., 60 cm or 100 cm) is suggested, as this may provide more reliable scaling and reduce errors, especially in challenging imaging conditions.
- Implementing a method that can detect reference objects independent of their color will allow the system to function more robustly across diverse environments and lighting conditions.
- The method of selecting landmark points and defining the connecting lines that represent body height should be optimized to ensure more precise measurements, especially in cases of complex or non-ideal postures.
- Integrating additional machine learning techniques to correct errors caused by varying lighting conditions, such as low-light or overexposed environments, will help maintain measurement accuracy even in suboptimal image conditions.

These enhancements will help create a more flexible and reliable system for infant height measurement, which can be further extended to different age groups and applications.

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